Bibliography

Classic Works

1. N. Bourbaki, *Eléments de Mathématique,* Book II of Part I, *Algèbre.* Paris: Hermann, 1942–58.
2. N. Jacobson, *Lectures in Abstract Algebra.* Princeton, N.J.: Van Nostrand, vols. I, 1951, II, 1953, and III, 1964.
3. O. Schreier and E. Sperner, *Introduction to Modern Algebra and Matrix Theory* (English translation), 2nd Ed. New York: Chelsea, 1959.
4. B. L. van der Waerden, *Modern Algebra* (English translation). New York: Ungar, vols. I, 1949, and II, 1950.

General Algebra Texts

5. M. Artin, *Algebra.* Englewood Cliffs, N.J.: Prentice-Hall, 1991.
6. A. A. Albert, *Fundamental Concepts of Higher Algebra.* Chicago: University of Chicago Press, 1956.
7. G. Birkhoff and S. MacLane, *A Survey of Modern Algebra,* 3rd Ed. New York: Macmillan, 1965.
8. J. A. Gallian, *Contemporary Abstract Algebra,* 2nd Ed. Lexington, Mass.: D. C. Heath, 1990.
9. I. N. Herstein, *Topics in Algebra.* New York: Blaisdell, 1964.
10. T. W. Hungerford, *Algebra.* New York: Springer, 1974.
11. S. Lang, *Algebra.* Reading, Mass.: Addison-Wesley, 1965.
12. S. MacLane and G. Birkhoff, *Algebra.* New York: Macmillan, 1967.
13. N. H. McCoy, *Introduction to Modern Algebra.* Boston: Allyn and Bacon, 1960.
14. G. D. Mostow, J. H. Sampson, and J. Meyer, *Fundamental Structures of Algebra.* New York: McGraw-Hill, 1963.
15. W. W. Sawyer, *A Concrete Approach to Abstract Algebra.* San Francisco: Freeman, 1959.

Group Theory

16. W. Burnside, *Theory of Groups of Finite Order,* 2nd Ed. New York: Dover, 1955.
17. H. S. M. Coxeter and W. O. Moser, *Generators and Relations for Discrete Groups,* 2nd Ed. Berlin: Springer, 1965.
18. M. Hall, Jr., *The Theory of Groups*. New York: Macmillan, 1959.
19. A. G. Kurosh, *The Theory of Groups* (English translation). New York: Chelsea, vols. I, 1955, and II, 1956.
20. W. Ledermann, *Introduction to the Theory of Finite Groups,* 4th rev. Ed. New York: Interscience, 1961.
21. J. G. Thompson and W. Feit, "Solvability of Groups of Odd Order." *Pac. J. Math.*, **13** (1963), 775–1029.
22. M. A. Rabin, "Recursive Unsolvability of Group Theoretic Problems." *Ann. Math.*, **67** (1958), 172–194.

Ring Theory

23. W. W. Adams and P. Loustaunau, *An Introduction to Gröbner Bases* (Graduate Studies in Mathematics, vol. 3). Providence, R.I.: American Mathematical Society, 1994.
24. E. Artin, C. J. Nesbitt, and R. M. Thrall, *Rings with Minimum Condition.* Ann Arbor: University of Michigan Press, 1944.
25. N. H. McCoy, *Rings and Ideals* (Carus Monograph No. 8). Buffalo: The Mathematical Association of America; LaSalle, III.: Open Court, 1948.
26. N. H. McCoy, *The Theory of Rings*. New York: Macmillan, 1964.

Field Theory

27. E. Artin, *Galois Theory* (Notre Dame Mathematical Lecture No. 2), 2nd Ed. Notre Dame, Ind.: University of Notre Dame Press, 1944.
28. O. Zariski and P. Samuel, *Commutative Algebra*. Princeton, N.J.: Van Nostrand, vol. I, 1958.

Number Theory

29. G. H. Hardy and E. M. Wright, *An Introduction to the Theory of Numbers,* 4th Ed. Oxford: Clarendon Press, 1960.
30. S. Lang, *Algebraic Numbers*. Reading, Mass.: Addison-Wesley, 1964.
31. W. J. LeVeque, *Elementary Theory of Numbers,* Reading, Mass.: Addison-Wesley, 1962.
32. W. J. LeVeque, *Topics in Number Theory*. Reading, Mass.: Addison-Wesley, 2 vols., 1956.
33. T. Nagell, *Introduction to Number Theory*. New York: Wiley, 1951.
34. I. Nivin and H. S. Zuckerman, *An Introduction to the Theory of Numbers*. New York: Wiley, 1960.
35. H. Pollard, *The Theory of Algebraic Numbers* (Carus Monograph No. 9). Buffalo: The Mathematical Association of America; New York: Wiley, 1950.
36. D. Shanks, *Solved and Unsolved Problems in Number Theory*. Washington, D.C.: Spartan Books, vol. I, 1962.
37. B. M. Stewart, *Theory of Numbers,* 2nd Ed. New York: Macmillan, 1964.
38. J. V. Uspensky and M. H. Heaslet, *Elementary Number Theory*. New York: McGraw-Hill, 1939.
39. E. Weiss, *Algebraic Number Theory*. New York: McGraw-Hill, 1963.

Homological Algebra

40. J. P. Jans, *Rings and Homology*. New York: Holt, 1964.
41. S. MacLane, *Homology*. Berlin: Springer, 1963.

Other References

42. A. A. Albert (ed.), *Studies in Modern Algebra* (MAA Studies in Mathematics, vol. 2). Buffalo: The Mathematical Association of America; Englewood Cliffs, N.J.: Prentice-Hall, 1963.
43. E. Artin, *Geometric Algebra*. New York: Interscience, 1957.
44. R. Courant and R. Robbins, *What Is Mathematics*? Oxford University Press, 1941.
45. H. S. M. Coxeter, *Introduction to Geometry,* 2nd Ed. New York: Wiley, 1969.
46. R. H. Crowell and R. H. Fox, *Introduction to Knot Theory*. New York: Ginn, 1963.
47. H. B. Edgerton, *Elements of Set Theory*. San Diego: Academic Press, 1977.
48. C. Schumacher, *Chapter Zero*. Reading, Mass.: Addison-Wesley, 1996.

Notations

$\in, a \in S$	membership, 1
$\varnothing$	empty set, 1
$\notin, a \notin S$	nonmembership, 1
$\{x \mid P(x)\}$	set of all x such that $P(x)$, 1
$B \subseteq A$	set inclusion, 2
$B \subset A$	subset $B \neq A$, 2
$A \times B$	Cartesian product of sets, 3
$\mathbb{Z}$	integers, 3
$\mathbb{Q}$	rational numbers, 3
$\mathbb{R}$	real numbers, 3
$\mathbb{C}$	complex numbers, 3
$\mathbb{Z}^+, \mathbb{Q}^+, \mathbb{R}^+$	positive elements of $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, 3
$\mathbb{Z}^*, \mathbb{Q}^*, \mathbb{R}^*, \mathbb{C}^*$	nonzero elements of $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$, 3
$\mathscr{R}$	relation, 3
$\lvert A \rvert$	number of elements in A, 4; as order of group, 50
$\phi : A \to B$	mapping of A into B by ϕ, 4
$\phi(a)$	image of element a under ϕ, 4
$\phi[A]$	image of set A under ϕ, 4
$\leftrightarrow$	one-to-one correspondence, 4
ϕ^{-1}	the inverse function of ϕ, 5
$\aleph_0$	cardinality of $\mathbb{Z}^+$, 5
$\bar{x}$	cell containing $x \in S$ in a partition of S, 6
$\equiv_n, a \equiv b (\mathrm{mod}\, n)$	congruence modulo n, 7
$\mathscr{P}(A)$	power set of A, 9
U	set of all $z \in \mathbb{C}$ such that $\lvert z \rvert = 1$, 15
$\mathbb{R}_c$	set of all $x \in \mathbb{R}$ such that $0 \leq x < c$, 16
$+_c$	addition modulo c, 16
U_n	group of nth roots of unity, 18

$\mathbb{Z}_n$	$\{0, 1, 2, \cdots, n-1\}$, 18
	cyclic group $\{0, 1, \cdots, n-1\}$ under addition modulo n, 54
	group of residue classes modulo n, 137
	ring $\{0, 1, \cdots, n-1\}$ under addition and multiplication modulo n, 169
$*, a * b$	binary operation, 20
$\circ, f \circ g, \sigma\tau$	function composition, 22, 76
$\langle S, * \rangle$	binary structure, 29
$\simeq, S \simeq S'$	isomorphic structures, 30
e	identity element, 32
$M_{m\times n}(S)$	$m \times n$ matrices with entries from S, 40, 477
$M_n(S)$	$n \times n$ matrices with entries from S, 40, 477
$\mathrm{GL}(n, \mathbb{R})$	general linear group of degree n, 40
$\det(A)$	determinant of square matrix A, 46, 479
$a^{-1}, -a$	inverse of a, 49
$H \leq G; K \leq L$	subgroup inclusion, 50; substructure inclusion, 173
$H < G; K < L$	subgroup $H \neq G$, 50; substructure $K \neq L$, 173
$\langle a \rangle$	cyclic subgroup generated by a, 54
	principal ideal generated by a, 250
$n\mathbb{Z}$	subgroup of $\mathbb{Z}$ generated by n, 54
	subring (ideal) of $\mathbb{Z}$ generated by n, 169, 250
gcd	greatest common divisor, 62, 258, 395
$\cap_{i\in I} S_i$, $S_1 \cap S_2 \cap \cdots \cap S_n$	intersection of sets, 69
S_A	group of permutations of A, 77
ι	identity map, 77
S_n	symmetric group on n letters, 78
$n!$	n factorial, 78
D_n	nth dihedral group, 79
A_n	alternating group on n letters, 93
$aH, a + H$	left coset of H containing a, 97
$Ha, H + a$	right coset of H containing a, 97
$(G : H)$	index of H in G, 101
φ	Euler phi-function, 104, 187
$\prod_{i=1}^n S_i$, $S_1 \times S_2 \times \cdots \times S_n$	Cartesian product of sets, 104
$\prod_{i=1}^n G_i$	direct product of groups, 104, 105
$\oplus_{i=1}^n G_i$	direct sum of groups, 105
lcm	least common multiple, 107
$\overline{G}_i$	natural subgroup of $\prod_{i=1}^n G_i$, 107
ϕ_c	evaluation homomorphism, 126
π_i	projection onto ith component, 127
$\phi^{-1}[B]$	inverse image of the set B under ϕ, 128
$\mathrm{Ker}(\phi)$	kernel of homomorphism ϕ, 129
$G/N; R/N$	factor group, 137; factor ring, 242
γ	canonical residue class map, 139, 140
i_g	inner automorphism, 141
$Z(G)$	center of the group G, 150
C	commutator subgroup, 150
X_g	subset of elements of X left fixed by g, 157

G_x	isotropy subgroup of elements of G leaving x fixed, 157
Gx	orbit of x under G, 158
$R[x]$	polynomial ring with coefficients in R, 200
$F(x)$	field of quotients of $F[x]$, 201
$\mathrm{F}(x_1, \cdots, x_n)$	field of rational functions in n indeterminates, 201
$\Phi_p(x)$	cyclotomic polynomial of degree $p-1$, 216, 217
$\mathrm{End}(A)$	endomorphisms of A, 221
RG	group ring, 223
FG	group algebra over the field F, 223
$\mathbb{H}$	quaternions, 224, 225
$R[[x]]$	formal power series ring in x over R, 231
$F((x))$	formal Laurent series field in x over F, 231
$F[\mathbf{x}]$	ring of polynomials in $x_1, \cdots, x_n$ over F, 255
$V(S)$	algebraic variety of polynomials in S, 255
$\langle b_1, \cdots, b_r \rangle$	ideal generated by elements $b_1, \cdots, b_r$, 255
$\mathrm{lt}(f)$	leading term of the polynomial f, 260
$\mathrm{lp}(f)$	power product of $\mathrm{lt}(f)$, 260
$\mathrm{irr}(\alpha, F)$	irreducible polynomial for α over F, 269
$\deg(\alpha, F)$	degree of α over F, 269
$F(\alpha)$	field obtained by adjoining α to field F, 270
$[E : F]$	degree of E over F, 283
$F(\alpha_1, \cdots, \alpha_n)$	field obtained by adjoining $\alpha_1, \cdots, \alpha_n$ to F, 285
$\overline{F_E}$	algebraic closure of F in E, 286
$\overline{F}$	an algebraic closure of F, 287, 288
$\mathrm{GF}(p^n)$	Galois field of order p^n, 300
HN	product set, 308
$H \vee N$	subgroup join, 308
$N[H]$	normalizer of H, 323
$F[A]$	free group on A, 341, 342
$(x_j : r_i)$	group presentation, 348
∂_n	boundary homomorphism, 357
$C_n(X)$	n-chains of X, 358
$Z_n(X)$	n-cycles of X, 359
$B_n(X)$	n-boundaries of X, 359
$H_n(X)$	nth homology group of X, 361
$\delta^{(n)}$	coboundary homomorphism, 363
$C^{(n)}(X)$	n-cochains of X, 363
$Z^{(n)}(X)$	n-cocycles of X, 363
$H^{(n)}(X)$	n-coboundaries of X, 363
$H^{(n)}(X)$	nth cohomology group of X, 363
S^n	n-sphere, 364
E^n	n-cell or n-ball, 364
$\chi(X)$	Euler characteristic of X, 374
f_{*n}	homology homomorphism induced from $f : X \to Y$, 375, 381
$\langle A, \partial \rangle$	chain complex, 381
$\bar{\partial}_k$	relative boundary operator, 382
$H_k(A/A')$	kth relative homology group of chain complex A modulo A', 383
$H_k(X, Y)$	kth relative homology of simplicial complex X modulo Y, 383
$a \mid b$	a divides (is a factor of) b, 389
UFD	unique factorization domain, 390

PID	principal ideal domain, 391
$\cup_{i\in I} S_i$, $S_1 \cup S_2 \cup \cdots \cup S_n$	union of sets, 391
ν	Euclidean norm, 401
$N(\alpha)$	norm of α, 408, 410, 455
$\psi_{\alpha,\beta}$	conjugation isomorphism of $F(\alpha)$ with $F(\beta)$, 416
$E_{\{\sigma_i\}}$, E_H	subfield of E left fixed by all σ_i or all $\sigma \in H$, 419
$G(E/F)$	automorphism group of E over F, 420
$\{E : F\}$	index of E over F, 428

Answers to Odd-Numbered Exercises Not Asking for Definitions or Proofs

SECTION 0

1. $\{-\sqrt{3}, \sqrt{3}\}$

3. $\{1, -1, 2, -2, 3, -3, 4, -4, 5, -5, 6, -6, 10, -10, 12, -12, 15, -15, 20, -20, 30, -30, 60, -60\}$

5. Not a set (not well defined). A case can also be made for the empty set $\varnothing$.

7. The set $\varnothing$

9. The set $\mathbb{Q}$

11. $(a, 1), (a, 2), (a, c), (b, 1), (b, 2), (b, c), (c, 1), (c, 2), (c, c)$

13. Draw the line through P and x, and let y be the point where it intersects the line segment CD.

17. Conjecture: $n(\mathscr{P}(A)) = 2^s$. (Proofs are usually omitted from answers.)

21. $10^2, 10^5, 10^{\aleph_0} = 12^{\aleph_0} = 2^{\aleph_0} = |\mathbb{R}|$. (The numbers x where $0 \leq x \leq 1$ can be written to base 12 and to base 2 as well as to base 10.)

23. 1 **25.** 5 **27.** 52

29. Not an equivalence relation

31. An equivalence relation; $\overline{0} = \{0\}, \overline{a} = \{a, -a\}$ for each nonzero $a \in \mathbb{R}$

33. An equivalence relation;
$\overline{1} = \{1, 2, \cdots, 9\}$,
$\overline{10} = \{10, 11, \cdots, 99\}$,
$\overline{100} = \{100, 101, \cdots, 999\}$, and in general
$\overline{10^n} = \{10^n, 10^n + 1, \cdots, 10^{n+1} - 1\}$

35. **i.** $\{1, 3, 5, \cdots\}, \{2, 4, 6, \cdots\}$
ii. $\{1, 4, 7, \cdots\}, \{2, 5, 8, \cdots\}, \{3, 6, 9, \cdots\}$
iii. $\{1, 6, 11, \cdots\}, \{2, 7, 12, \cdots\}, \{3, 8, 13, \cdots\}, \{4, 9, 14, \cdots\}, \{5, 10, 15, \cdots\}$

37. The name *two-to-two function* suggests that such a function f should carry every pair of distinct points into two distinct points. Such a function is one to one in the conventional sense. (If the domain has only one element, a function cannot fail to be two to two, since the only way it can fail to be two to two is to carry two points into one point, and the set does not have two points.) Conversely, every function that is one to one in the conventional sense carries any pair of points into two distinct points. Thus the functions conventionally called one to one are precisely those that carry two points into two points, which is a much more intuitive unidirectional way of regarding them. Also, the standard way of trying

to show a function is one to one is precisely to show that it does not carry two points into just one point. Thus, proving a function is one to one becomes more natural in the two-to-two terminology.

SECTION 1

1. $-i$ **3.** $-i$ **5.** $23 + 7i$
7. $17 - 15i$ **9.** $-4 + 4i$
11. $2\sqrt{13}$ **13.** $\sqrt{2}\left(-\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}i\right)$
15. $\sqrt{34}\left(\frac{-3}{\sqrt{34}} + \frac{5}{\sqrt{34}}i\right)$
17. $\frac{1}{\sqrt{2}} \pm \frac{1}{\sqrt{2}}i, -\frac{1}{\sqrt{2}} \pm \frac{1}{\sqrt{2}}i$ **19.** $3i, \pm\frac{3\sqrt{3}}{2} - \frac{3}{2}i$
21. $\sqrt{3} \pm i, \pm 2i, -\sqrt{3} \pm i$ **23.** 4 **25.** $\frac{3}{8}$ **27.** $\sqrt{2}$
29. 11 **31.** 5 **33.** 1, 7
35. $\zeta^0 \leftrightarrow 0, \zeta^3 \leftrightarrow 7, \zeta^4 \leftrightarrow 4, \zeta^5 \leftrightarrow 1, \zeta^6 \leftrightarrow 6, \zeta^7 \leftrightarrow 3$
37. With $\zeta \leftrightarrow 4$, we must have $\zeta^2 \leftrightarrow 2$, $\zeta^3 \leftrightarrow 0$, and $\zeta^4 \leftrightarrow 4$ again, which is impossible for a one-to-one correspondence.
39. Multiplying, we obtain

$$z_1 z_2 = |z_1||z_2|[(\cos\theta_1 \cos\theta_2 - \sin\theta_1 \sin\theta_2) + (\cos\theta_1 \sin\theta_2 + \sin\theta_1 \cos\theta_2)i]$$

and the desired result follows at once from Exercise 38 and the equation $|z_1||z_2| = |z_1 z_2|$.

SECTION 2

1. e, b, a **3.** a, c. $*$ is not associative.
5. Top row: d; second row: a; fourth row: c, b.
7. Not commutative, not associative
9. Commutative, associative
11. Not commutative, not associative
13. 8, 729, $n^{[n(n+1)/2]}$
17. No. Condition 2 is violated. **19.** Yes
21. No. Condition 1 is violated.
23. **a.** Yes. **b.** Yes
25. Let $S = \{?, \Delta\}$. Define $*$ and $*'$ on S by $a * b = ?$ and $a *' b = \Delta$ for all $a, b \in S$. (Other answers are possible.)
27. True **29.** True
31. False. Let $f(x) = x^2$, $g(x) = x$, and $h(x) = 2x + 1$. Then
$(f(x) - g(x)) - h(x) = x^2 - 3x - 1$ but
$f(x) - (g(x) - h(x)) = x^2 - (-x - 1) = x^2 + x + 1$.
33. True **35.** False. Let $*$ be $+$ and let $*'$ be on $\mathbb{Z}$.

SECTION 3

1. **i.** ϕ must be one to one. **ii.** $\phi[S]$ must be all of S'.
iii. $\phi(a * b) = \phi(a) *' \phi(b)$ for all $a, b \in S$.
3. No, because ϕ does not map $\mathbb{Z}$ onto $\mathbb{Z}'$. $\phi(n) \neq 1$ for all $n \in \mathbb{Z}$.
5. Yes. **7.** Yes **9.** Yes
11. No, because $\phi(x^2) = \phi(x^2 + 1)$.
13. No, because $\phi(f) = x + 1$ has no solution $f \in F$.

15. No, because $\phi(f) = 1$ has no solution $f \in F$.

17. **a.** $m * n = mn - m - n + 2$; identity element 2

b. $m * n = mn + m + n$; identity element 0

19. **a.** $a * b = \frac{1}{3}(ab + a + b - 2)$; identity element 2

b. $a * b = 3ab - a - b + \frac{2}{3}$; identity element $\frac{2}{3}$

25. No. If $\langle S, * \rangle$ has a left identity element e_L and a right identity element e_R, then $e_L = e_R$. (It is our practice to omit proofs from answers.)

SECTION 4

1. No. $\mathscr{G}_3$ fails. **3.** No. $\mathscr{G}_1$ fails. **5.** No. $\mathscr{G}_1$ fails.

7. The group $\langle U_{1000}, \cdot \rangle$ of solutions of $z^{1000} = 1$ in $\mathbb{C}$ under multiplication has 1000 elements.

9. An equation of the form $x * x * x * x = e$ has four solutions in $\langle U, \cdot \rangle$, one solution in $\langle \mathbb{R}, + \rangle$, and two solutions in $\langle \mathbb{R}^*, \cdot \rangle$.

11. Yes **13.** Yes

15. No. The matrix with all entries 0 is upper triangular, but has no inverse.

17. Yes.

19. (Proofs are omitted.) **c.** $-1/3$

21. 2, 3. (It gets harder for 4 elements, where the answer is *not* 4.)

25. **a.** F **c.** T **e.** F **g.** T **i.** F

SECTION 5

1. Yes **3.** Yes **5.** Yes **7.** $\mathbb{Q}^+$ and $\{\pi^n \mid n \in \mathbb{Z}\}$ **9.** Yes

11. No. Not closed under multiplication.

13. Yes

15. **a.** Yes **b.** No. It is not even a subset of $\tilde{\mathrm{F}}$.

17. **a.** No. Not closed under addition. **b.** Yes

19. **a.** Yes **b.** No. The zero constant function is not in $\tilde{\mathrm{F}}$.

21. **a.** $-50, -25, 0, 25, 50$ **b.** $4, 2, 1, 1/2, 1/4$ **c.** $1, \pi, \pi^2, 1/\pi, 1/\pi^2$

23. All matrices $\begin{bmatrix} 1 & n \\ 0 & 1 \end{bmatrix}$ for $n \in \mathbb{Z}$

25. All matrices of the form $\begin{bmatrix} 4^n & 0 \\ 0 & 4^n \end{bmatrix}$ or $\begin{bmatrix} 0 & -2^{2n+1} \\ -2^{2n+1} & 0 \end{bmatrix}$ for $n \in \mathbb{Z}$

27. 4 **29.** 3 **31.** 4 **33.** 2 **35.** 3

39. **a.** T **c.** T **e.** F **g.** F **i.** T

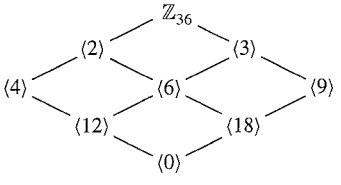
SECTION 6

1. $q = 4, r = 6$ **3.** $q = -7, r = 6$ **5.** 8 **7.** 60

9. 4 **11.** 16 **13.** 2 **15.** 2 **17.** 6 **19.** 4

21. An infinite cyclic group

23.



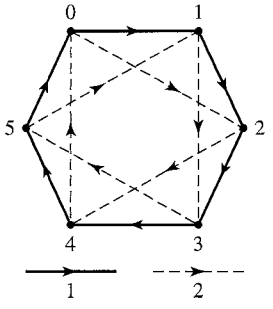
25. 1, 2, 3, 6 **27.** 1, 2, 3, 4, 6, 12 **29.** 1, 17
33. The Klein 4-group **35.** $\mathbb{Z}_2$ **37.** $\mathbb{Z}_8$
39. $\frac{1}{2}(1+i\sqrt{3})$ and $\frac{1}{2}(1-i\sqrt{3})$
41. $\frac{1}{2}(\sqrt{3}+i)$, $\frac{1}{2}(\sqrt{3}-i)$, $\frac{1}{2}(-\sqrt{3}+i)$, $\frac{1}{2}(-\sqrt{3}-i)$
51. $(p-1)(q-1)$

SECTION 7

1. 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11 **3.** 0, 2, 4, 6, 8, 10, 12, 14, 16
5. $\cdots, -24, -18, -12, -6, 0, 6, 12, 18, 24, \cdots$
7. **a.** a^3b **b.** a^2 **c.** a^2
9.

	e	a	b	c	d	f
e	e	a	b	c	d	f
a	a	e	c	b	f	d
b	b	d	e	f	a	c
c	e	f	a	d	e	b
d	d	b	f	e	c	a
f	f	c	d	a	b	e

11. Choose a pair of generating directed arcs, call them *arc1* and *arc2*, start at any vertex of the digraph, and see if the sequences *arc1*, *arc2* and *arc2*, *arc1* lead to the same vertex. (This corresponds to asking if the two corresponding group generators commute.) The group is commutative if and only if these two sequences lead to the same vertex for *every pair* of generating directed arcs.
13. It is not obvious, since a digraph of a cyclic group might be formed using a generating set of two or more elements, no one of which generates the group.
15.



17. **a.** Starting from any vertex a, every path through the graph that terminates at that same vertex a represents a product of generators or their inverses that is equal to the identity and thus gives a relation.
b. $a^4 = e, b^2 = e, (ab)^2 = e$

SECTION 8

1. $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 1 & 2 & 3 & 6 & 5 & 4 \end{pmatrix}$ **3.** $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 3 & 4 & 1 & 6 & 2 & 5 \end{pmatrix}$

5. $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 6 & 1 & 5 & 4 & 3 \end{pmatrix}$

7. 2 **9.** ι

11. $\{1, 2, 3, 4, 5, 6\}$ **13.** $\{1, 5\}$

15. $\epsilon, \rho, \rho^2, \rho^3, \phi, \rho\phi, \rho^2\phi, \rho^3\phi$ where their ϕ is our μ_1. This gives our elements in the order $\rho_0, \rho_1, \rho_2, \rho_3, \mu_1, \delta_1, \mu_2, \delta_2$.

17. 24

19. Referring to Table 8.12, we find that $\langle\rho_0\rangle = \{\rho_0\}$, $\langle\rho_1\rangle = \langle\rho_3\rangle = \{\rho_0, \rho_1, \rho_2, \rho_3\}$, $\langle\rho_2\rangle = \{\rho_0, \rho_2\}$, $\langle\mu_1\rangle = \{\rho_0, \mu_1\}$, $\langle\mu_2\rangle = \{\rho_0, \mu_2\}$, $\langle\delta_1\rangle = \{\rho_0, \delta_1\}$, and $\langle\delta_2\rangle = \{\rho_0, \delta_2\}$. These are all the cyclic subgroups. A subgroup containing one of the "turn the square over" permutations μ_1, μ_2, δ_1, or δ_2 and also containing ρ_1 or ρ_3 will describe all positions of the square so it must be the entire group D_4. Checking the line of the table opposite μ_1, we see that the only other elements that can be in a proper subgroup with μ_1 are ρ_2, μ_2, and, of course, ρ_0. We check that $\{\rho_0, \rho_2, \mu_1, \mu_2\}$ is closed under multiplication and is a subgroup. Checking the row of the table opposite μ_2 gives the same subgroup. Checking the rows opposite δ_1 and opposite δ_2 gives the subgroup $\{\rho_0, \rho_2, \delta_1, \delta_2\}$ as the only remaining possibility, using the same reasoning.

21. **a.** These are "elementary permutation matrices," resulting from permuting the rows of the identity matrix. When another matrix A is multiplied on the left by one of these matrices P, the rows of A are permuted in the same fashion that the rows of the 3×3 identity matrix were permuted to obtain P. Because all 6 possible permutations of the three rows are present, we see they will act just like the elements of S_3 in permuting the entries 1, 2, 3 of the given column vector. Thus they form a group because S_3 is a group.

b. The symmetric group S_3.

23. $\mathbb{Z}_2$ **25.** D_4

27. For $\mathbb{Z}_4$, $\lambda_0 = \begin{pmatrix} 0 & 1 & 2 & 3 \\ 0 & 1 & 2 & 3 \end{pmatrix}$, $\lambda_1 = \begin{pmatrix} 0 & 1 & 2 & 3 \\ 1 & 2 & 3 & 0 \end{pmatrix}$, $\lambda_2 = \begin{pmatrix} 0 & 1 & 2 & 3 \\ 2 & 3 & 0 & 1 \end{pmatrix}$, $\lambda_3 = \begin{pmatrix} 0 & 1 & 2 & 3 \\ 3 & 0 & 1 & 2 \end{pmatrix}$. The table for the left regular representation is the same as the table for $\mathbb{Z}_4$ with n replaced by λ_n. For S_3, $\rho_0 = \begin{pmatrix} r_0 & r_1 & r_2 & m_1 & m_2 & m_3 \\ r_0 & r_1 & r_2 & m_1 & m_2 & m_3 \end{pmatrix}$, $\rho_1 = \begin{pmatrix} r_0 & r_1 & r_2 & m_1 & m_2 & m_3 \\ r_1 & r_2 & r_0 & m_2 & m_3 & m_1 \end{pmatrix}$, etc., where the bottom row in the permutation ρ_σ consists of the elements of S_3 in the order they appear down the column under σ in Table 8.8. The table for this right regular representation is the same as the table for S_3 with σ replaced by ρ_σ.

31. Not a permutation **33.** Not a permutation

35. **a.** T **c.** T **e.** T **g.** F **i.** F

37. A monoid **41.** No **43.** Yes

SECTION 9

1. $\{1, 2, 5\}, \{3\}, \{4, 6\}$

3. $\{1, 2, 3, 4, 5\}, \{6\}, \{7, 8\}$

5. $\{2n \mid n \in \mathbb{Z}\}, \{2n + 1 \mid n \in \mathbb{Z}\}$

7. $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 4 & 1 & 3 & 5 & 8 & 6 & 2 & 7 \end{pmatrix}$

9. $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 5 & 4 & 3 & 7 & 8 & 6 & 2 & 1 \end{pmatrix}$

11. $(1, 3, 4)(2, 6)(5, 8, 7) = (1, 4)(1, 3)(2, 6)(5, 7)(5, 8)$

13. **a.** 4

b. A cycle of length n has order n.

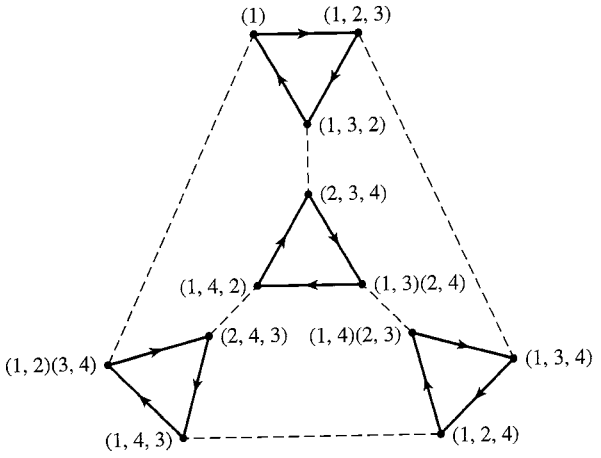
c. σ has order 6; τ has order 4.

d. 6 in Exercises 10 and 11, 8 in Exercise 12.

e. The order of a permutation expressed as a product of disjoint cycles is the least common multiple of the lengths of the cycles.

15. 6 **17.** 30

19.



23. **a.** F **c.** F **e.** F **g.** T **i.** T

SECTION 10

1. $4\mathbb{Z} = \{\cdots, -8, -4, 0, 4, 8, \cdots\}$
$1 + 4\mathbb{Z} = \{\cdots, -7, -3, 1, 5, 9, \cdots\}$
$2 + 4\mathbb{Z} = \{\cdots, -6, -2, 2, 6, 10, \cdots\}$
$3 + 4\mathbb{Z} = \{\cdots, -5, -1, 3, 7, 11, \cdots\}$

3. $\langle 2 \rangle = \{0, 2, 4, 6, 8, 10\}$, $1 + \langle 2 \rangle = \{1, 3, 5, 7, 9, 11\}$

5. $\langle 18 \rangle = \{0, 18\}$, $1 + \langle 18 \rangle = \{1, 19\}$, $2 + \langle 18 \rangle = \{2, 20\}, \cdots, 17 + \langle 18 \rangle = \{17, 35\}$

7. $\{\rho_0, \mu_2\}, \{\rho_1, \delta_1\}, \{\rho_2, \mu_1\}, \{\rho_3, \delta_2\}$. Not the same.

9. $\{\rho_0, \rho_2\}, \{\rho_1, \rho_3\}, \{\mu_1, \mu_2\}, \{\delta_1, \delta_2\}$

11. Yes, we get a coset group isomorphic to the Klein 4-group V.

	ρ_0	ρ_2	ρ_1	ρ_3	μ_1	μ_2	δ_1	δ_2
ρ_0	ρ_0	ρ_2	ρ_1	ρ_3	μ_1	μ_2	δ_1	δ_2
ρ_2	ρ_2	ρ_0	ρ_3	ρ_1	μ_2	μ_1	δ_2	δ_1
ρ_1	ρ_1	ρ_3	ρ_2	ρ_0	δ_1	δ_2	μ_2	μ_1
ρ_3	ρ_3	ρ_1	ρ_0	ρ_2	δ_2	δ_1	μ_1	μ_2
μ_1	μ_1	μ_2	δ_2	δ_1	ρ_0	ρ_2	ρ_3	ρ_1
μ_2	μ_2	μ_1	δ_1	δ_2	ρ_2	ρ_0	ρ_1	ρ_3
δ_1	δ_1	δ_2	μ_1	μ_2	ρ_1	ρ_3	ρ_0	ρ_2
δ_2	δ_2	δ_1	μ_2	μ_1	ρ_3	ρ_1	ρ_2	ρ_0

13. 3 **15.** 24

19. **a.** T **c.** T **e.** T **g.** T **i.** F

21. $G = \mathbb{Z}_2$, subgroup $H = \mathbb{Z}_2$.

23. Impossible. The number of cells must divide the order of the group, and 12 does not divide 6.

SECTION 11

1.

Element	*Order*	*Element*	*Order*
(0, 0)	1	(0, 2)	2
(1, 0)	2	(1, 2)	2
(0, 1)	4	(0, 3)	4
(1, 1)	4	(1, 3)	4

The group is not cyclic

3. 2 **5.** 9 **7.** 60

9. $\{(0, 0), (0, 1)\}$, $\{(0, 0), (1, 0)\}$, $\{(0, 0), (1, 1)\}$

11. $\{(0, 0), (0, 1), (0, 2), (0, 3)\}$
$\{(0, 0), (0, 2), (1, 0), (1, 2)\}$
$\{(0, 0), (1, 1), (0, 2), (1, 3)\}$

13. $\mathbb{Z}_{20} \times \mathbb{Z}_3$, $\mathbb{Z}_{15} \times \mathbb{Z}_4$, $\mathbb{Z}_{12} \times \mathbb{Z}_5$, $\mathbb{Z}_5 \times \mathbb{Z}_3 \times \mathbb{Z}_4$

15. 12

17. 120

19. 180

21. $\mathbb{Z}_8$, $\mathbb{Z}_2 \times \mathbb{Z}_4$, $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$

23. $\mathbb{Z}_{32}$, $\mathbb{Z}_2 \times \mathbb{Z}_{16}$, $\mathbb{Z}_4 \times \mathbb{Z}_8$, $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_8$, $\mathbb{Z}_2 \times \mathbb{Z}_4 \times \mathbb{Z}_4$,
$\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_4$, $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$

25. $\mathbb{Z}_9 \times \mathbb{Z}_{121}$, $\mathbb{Z}_3 \times \mathbb{Z}_3 \times \mathbb{Z}_{121}$, $\mathbb{Z}_9 \times \mathbb{Z}_{11} \times \mathbb{Z}_{11}$, $\mathbb{Z}_3 \times \mathbb{Z}_3 \times \mathbb{Z}_{11} \times \mathbb{Z}_{11}$

29. a.

n	2	3	4	5	6	7	8
number of groups	2	3	5	7	11	15	22

b. **i)** 225 **ii)** 225 **iii)** 110

31. a. It is abelian when the arrows on both n-gons have the same (clockwise or counterclockwise) direction.
b. $\mathbb{Z}_2 \times \mathbb{Z}_n$
c. When n is odd.
d. The dihedral group D_n.

33. $\mathbb{Z}_2$ is an example.

35. S_3 is an example.

37. The numbers are the same. **41.** $\{-1, 1\}$

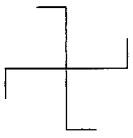
SECTION 12

1. a. The only isometries of $\mathbb{R}$ leaving a number c fixed are the reflection through c that carries $c + x$ to $c - x$ for all $x \in \mathbb{R}$, and the identity map.
b. The isometries of $\mathbb{R}^2$ that leave a point P fixed are the rotations about P through any angle θ where $0 \leq \theta < 360°$ and the reflections across any axis that passes through P.
c. The only isometries of $\mathbb{R}$ that carry a line segment into itself are the reflection through the midpoint of the line segment (see the answer to part (a)) and the identity map.
d. The isometries of $\mathbb{R}^2$ that carry a line segment into itself are a rotation of 180° about the midpoint of the line segment, a reflection in the axis containing the line segment, a reflection in the axis perpendicular to the line segment at its midpoint, and the identity map.
e. The isometries of $\mathbb{R}^3$ that carry a line segment into itself include rotations through any angle about an axis that contains the line segment, reflections across any plane that contains the line segment, and reflection across the plane perpendicular to the line segment at its midpoint.

3.

	τ	ρ	μ	γ
τ	τ	ρ	$\mu\gamma$	$\mu\gamma$
ρ	ρ	$\rho\tau$	$\mu\gamma$	$\mu\gamma$
μ	$\mu\gamma$	$\mu\gamma$	$\tau\rho$	$\tau\rho$
γ	$\mu\gamma$	$\mu\gamma$	$\tau\rho$	$\tau\rho$

5. 

7. 

9. *Translation:* order ∞
Rotation: order any $n \geq 2$ or ∞
Reflection: order 2
Glide reflection: order ∞

11. Rotations **13.** Only the identity and reflections.

17. Yes. The product of two translations is a translation and the inverse of a translation is a translation.

19. Yes. There is only one reflection μ across one particular line L, and μ^2 is the identity, so we have a group isomorphic to $\mathbb{Z}_2$.

21. Only reflections and rotations (and the identity) because translations and glide reflections do not have finite order in the group of all plane isometries.

25. **a.** No **b.** No **c.** Yes **d.** No **e.** D_∞

27. **a.** Yes **b.** No **c.** No **d.** No **e.** D_∞

29. **a.** No **b.** No **c.** No **d.** Yes **e.** $\mathbb{Z}$

31. **a.** Yes. 90°, 180° **b.** Yes **c.** No

33. **a.** No **b.** No **c.** No

35. **a.** Yes. 180° **b.** Yes **c.** No

37. **a.** Yes. 120° **b.** Yes **c.** No

39. **a.** Yes. 90°, 180° **b.** Yes **c.** No **d.** $(-1, 1)$ and $(1, 1)$

41. **a.** Yes. 120° **b.** Yes **c.** No **d.** $(0, 1)$ and $(1, \sqrt{3})$

SECTION 13

1. Yes **3.** Yes **5.** No

7. Yes **9.** Yes

11. Yes **13.** Yes **15.** No

17. $\text{Ker}(\phi) = 7\mathbb{Z}; \phi(25) = 2$

19. $\text{Ker}(\phi) = 6\mathbb{Z}; \phi(20) = (1, 2, 7)(4, 5, 6)$

21. $\text{Ker}(\phi) = \{0, 4, 8, 12, 16, 20\}; \phi(14) = (1, 6)(4, 7)$

23. $\text{Ker}(\phi) = \{(0, 0)\}; \phi(4, 6) = (2, 18)$

25. 2 **27.** 2 **29.** For all $g \in G$

33. No nontrivial homomorphism. By Theorem 13.12, the image of ϕ would have to be a subgroup of $\mathbb{Z}_5$, and hence all of $\mathbb{Z}_5$ for a nontrivial ϕ. But the number of cosets of a subgroup of a finite group is a divisor of the order of the group, and 5 does not divide 12.

35. Let $\phi(m, n) = (m, 0)$ for $(m, n) \in \mathbb{Z}_2 \times \mathbb{Z}_4$.
37. Let $\phi(n) = \rho_n$ for $n \in \mathbb{Z}_3$, using our notation in the text for elements of S_3.
39. Let $\phi(m, n) = 2m$.
41. Viewing D_4 as a group of permutations, let $\phi(\sigma) = (1, 2)$ for odd $\sigma \in D_4$ and $\phi(\sigma)$ be the identity for even $\sigma \in D_4$.
43. Let $\phi(\sigma) = (1, 2)$ for odd $\sigma \in S_4$ and $\phi(\sigma)$ be the identity element for even $\sigma \in S_4$.
51. The image of ϕ is $\langle a \rangle$, and $\text{Ker}(\phi)$ must be some subgroup $n\mathbb{Z}$ of $\mathbb{Z}$.
53. $hk = kh$ **55.** h^n must be the identity e of G.

SECTION 14

1. 3 **3.** 4 **5.** 2 **7.** 2
9. 4 **11.** 3 **13.** 4 **15.** 1
21. **a.** When working with a factor group G/H, you would let a and b be elements of G, not elements of G/H. The student probably does not understand what elements of G/H look like and can write nothing sensible concerning them.
b. We must show that G/H is abelian. Let aH and bH be two elements of G/H.
23. **a.** T **c.** T **e.** T **g.** T **i.** T
29. $\{\rho_0, \mu_1\}$, $\{\rho_0, \mu_2\}$, and $\{\rho_0, \mu_3\}$
35. Example: Let $G = N = S_3$, and let $H = \{\rho_0, \mu_1\}$. Then N is normal in G, but $H \cap N = H$ is not normal in G.

SECTION 15

1. $\mathbb{Z}_2$ **3.** $\mathbb{Z}_4$ **5.** $\mathbb{Z}_4 \times \mathbb{Z}_8$ **7.** $\mathbb{Z}$ **9.** $\mathbb{Z}_3 \times \mathbb{Z} \times \mathbb{Z}_4$
11. $\mathbb{Z}_2 \times \mathbb{Z}$ **13.** $\mathbb{Z}(D_4) = C = \{\rho_0, \rho_2\}$
15. $Z(S_3 \times D_4) = \{(\rho_0, \rho_0), (\rho_0, \rho_2)\}$, using the notations for these groups in Section 8, $C = A_3 \times \{\rho_0, \rho_2\}$.
19. **a.** T **c.** F **e.** F **g.** F **i.** T
21. $\{f \in F^* | f(0) = 1\}$
23. Yes. Let $f(x) = 1$ for $x \geq 0$ and $f(x) = -1$ for $x < 0$. Then $f(x) \cdot f(x) = 1$ for all x, so $f^2 \in K^*$ but f is not in K^*. Thus fK^* has order 2 in F^*/K^*.
25. U
27. The multiplicative group U of complex numbers of absolute value 1
29. Let $G = \mathbb{Z}_2 \times \mathbb{Z}_4$. Then $H = \langle (1, 0) \rangle$ is isomorphic to $K = \langle (0, 2) \rangle$, but G/H is isomorphic to $\mathbb{Z}_4$ while G/K is isomorphic to $\mathbb{Z}_2 \times \mathbb{Z}_2$.
31. **a.** $\{e\}$ **b.** The whole group

SECTION 16

1. $X_{\rho_0} = X$, $X_{\rho_1} = \{C\}$, $X_{\rho_2} = \{m_1, m_2, d_1, d_2, C\}$, $X_{\rho_3} = \{C\}$,
$X_{\mu_1} = \{s_1, s_3, m_1, m_2, C, P_1, P_3\}$, $X_{\mu_2} = \{s_2, s_4, m_1, m_2, C, P_2, P_4\}$,
$X_{\delta_1} = \{2, 4, d_1, d_2, C\}$, $X_{\delta_2} = \{1, 3, d_1, d_2, C\}$.
3. $\{1, 2, 3, 4\}$, $\{s_1, s_2, s_3, s_4\}$, $\{m_1, m_2\}$, $\{d_1, d_2\}$, $\{C\}$, $\{P_1, P_2, P_3, P_4\}$
7. A transitive G-set has just one orbit.
9. **a.** $\{s_1, s_2, s_3, s_4\}$ and $\{P_1, P_2, P_3, P_4\}$
13. **b.** The set of points on the circle with center at the origin and passing through P
c. The cyclic subgroup $\langle 2\pi \rangle$ of $G = \mathbb{R}$
17. **a.** $K = g_0 H g_0^{-1}$.
b. *Conjecture:* H and K should be conjugate subgroups of G.

19.

	X	Y		Z		
	a	a	b	a	b	c
0	a	a	b	a	b	c
1	a	b	a	b	c	a
2	a	a	b	c	a	b
3	a	b	a	a	b	c
4	a	a	b	b	c	a
5	a	b	a	c	a	b

There are four of them: X, Y, Z, and $\mathbb{Z}_6$.

SECTION 17

1. 5 **3.** 2 **5.** 11,712
7. **a.** 45 **b.** 231
9. **a.** 90 **b.** 6,246

SECTION 18

1. 0 **3.** 1 **5.** $(1, 6)$
7. Commutative ring, no unity, not a field
9. Commutative ring with unity, not a field
11. Commutative ring with unity, not a field
13. No. $\{ri | r \in \mathbb{R}\}$ is not closed under multiplication.
15. $(1, 1), (1, -1), (-1, 1), (-1, -1)$
17. All nonzero $q \in \mathbb{Q}$ **19.** 1, 3
21. Let $\mathbb{R} = \mathbb{Z}$ with unity 1 and $\mathrm{R}' = \mathbb{Z} \times \mathbb{Z}$ with unity $1' = (1, 1)$. Let $\phi : R \to R'$ be defined by $\phi(n) = (n, 0)$. Then $\phi(1) = (1, 0) \neq 1'$.
23. $\phi_1 : \mathbb{Z} \to \mathbb{Z}$ where $\phi_1(n) = 0$, $\phi_2 : \mathbb{Z} \to \mathbb{Z}$ where $\phi_2(n) = n$
25. $\phi_1 : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$ where $\phi_1(n, m) = 0$, $\phi_2 : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$ where $\phi_2(n, m) = n$
$\phi_3 : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$ where $\phi_3(n, m) = m$
27. The reasoning is not correct since a product $(X - I_3)(X + I_3)$ of two matrices may be the zero matrix 0 without having either matrix be 0. Counterexample:

$$\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}^2 = I_3.$$

31. $a = 2, b = 3$ in $\mathbb{Z}_6$
33. **a.** T **c.** F **e.** T **g.** T **i.** T

SECTION 19

1. 0, 3, 5, 8, 9, 11 **3.** No solutions **5.** 0 **7.** 0 **9.** 12
11. $a^4 + 2a^2b^2 + b^4$ **13.** $a^6 + 2a^3b^3 + b^6$
17. **a.** F **c.** F **e.** T **g.** F **i.** F

19. **1.** $\text{Det}(A) = 0$. **2.** The column vectors of A are dependent.
3. The row vectors of A are dependent. **4.** Zero is an eigenvalue of A.
5. A is not invertible.

SECTION 20

1. 3 or 5 **3.** Any of 3, 5, 6, 7, 10, 11, 12, or 14. **5.** 2

7.

$\varphi(1) = 1$	$\varphi(7) = 6$	$\varphi(13) = 12$	$\varphi(19) = 18$	$\varphi(25) = 20$
$\varphi(2) = 1$	$\varphi(8) = 4$	$\varphi(14) = 6$	$\varphi(20) = 8$	$\varphi(26) = 12$
$\varphi(3) = 2$	$\varphi(9) = 6$	$\varphi(15) = 8$	$\varphi(21) = 12$	$\varphi(27) = 18$
$\varphi(4) = 2$	$\varphi(10) = 4$	$\varphi(16) = 8$	$\varphi(22) = 10$	$\varphi(28) = 12$
$\varphi(5) = 4$	$\varphi(11) = 10$	$\varphi(17) = 16$	$\varphi(23) = 22$	$\varphi(29) = 28$
$\varphi(6) = 2$	$\varphi(12) = 4$	$\varphi(18) = 6$	$\varphi(24) = 8$	$\varphi(30) = 8$

9. $(p-1)(q-1)$ **11.** $1 + 4\mathbb{Z}, 3 + 4\mathbb{Z}$ **13.** No solutions
15. No solutions
17. $3 + 65\mathbb{Z}, 16 + 65\mathbb{Z}, 29 + 65\mathbb{Z}, 42 + 65\mathbb{Z}, 55 + 65\mathbb{Z}$
19. 1 **21.** 9
23. **a.** F **c.** T **e.** T **g.** F **i.** F

SECTION 21

1. $\{q_1 + q_2 i \mid q_1, q_2 \in \mathbb{Q}\}$
15. It is isomorphic to the ring D of all rational numbers that can be expressed as a quotient of integers with denominator some power of 2.
17. It runs into trouble when we try to prove the transitive property in the proof of Lemma 5.4.2, for multiplicative cancellation may not hold. For $R = \mathbb{Z}_6$ and $T = \{1, 2, 4\}$ we have $(1, 2) \sim (2, 4)$ since $(1)(4) = (2)(2) = 4$ and $(2, 4) \sim (2, 1)$ since $(2)(1) = (4)(2)$ in $\mathbb{Z}_6$. However, $(1, 2)$ is not equivalent to $(2, 1)$ because $(1)(1) \neq (2)(2)$ in $\mathbb{Z}_6$.

SECTION 22

1. $f(x) + g(x) = 2x^2 + 5$, $f(x)g(x) = 6x^2 + 4x + 6$
3. $f(x) + g(x) = 5x^2 + 5x + 1$, $f(x)g(x) = x^3 + 5x$
5. 16 **7.** 7 **9.** 2 **11.** 0 **13.** 2, 3 **15.** 0, 2, 4
17. 0, 1, 2, 3
21. $0, x - 5, 2x - 10, x^2 - 25, x^2 - 5x, x^4 - 5x^3$. (Other answers are possible.)
23. **a.** T **c.** T **e.** F **g.** T **i.** T
25. **a.** They are the units of D. **b.** $1, -1$ **c.** 1, 2, 3, 4, 5, 6
27. **b.** F **c.** $F[x]$ **31.** **a.** 4, 27 **b.** $\mathbb{Z}_2 \times \mathbb{Z}_2, \mathbb{Z}_3 \times \mathbb{Z}_3 \times \mathbb{Z}_3$

SECTION 23

1. $q(x) = x^4 + x^3 + x^2 + x - 2, r(x) = 4x + 3$
3. $q(x) = 6x^4 + 7x^3 + 2x^2 - x + 2, r(x) = 4$
5. 2, 3 **7.** 3, 10, 5, 11, 14, 7, 12, 6
9. $(x-1)(x+1)(x-2)(x+2)$
11. $(x-3)(x+3)(2x+3)$

13. Yes. It is of degree 3 with no zeros in $\mathbb{Z}_5$.
$2x^3 + x^2 + 2x + 2$
15. *Partial answer:* $g(x)$ is irreducible over $\mathbb{R}$, but it is not irreducible over $\mathbb{C}$.
19. Yes. $p = 3$ **21.** Yes. $p = 5$
25. **a.** T **c.** T **e.** T **g.** T **i.** T
27. $x^2 + x + 1$
29. $x^2 + 1, x^2 + x + 2, x^2 + 2x + 2, 2x^2 + 2, 2x^2 + x + 1, 2x^2 + 2x + 1$
31. $p(p-1)^2/2$

SECTION 24

1. $1e + 0a + 3b$ **3.** $2e + 2a + 2b$ **5.** j **7.** $(1/50)j - (3/50)k$
9. $\mathbb{R}^*$, that is, $\{a_1 + 0i + 0j + 0k \mid a_1 \in \mathbb{R}, a_1 \neq 0\}$
11. **a.** F **c.** F **e.** F **g.** T **i.** T
c. If $|A| = 1$, then $\text{End}(A) = \{0\}$. **e.** $0 \in \text{End}(A)$ is not in $\text{Iso}(A)$.
19. **a.** $K = \begin{bmatrix} i & 0 \\ 0 & -i \end{bmatrix}$.
b. Denoting by B the matrix with coefficient b and by C the matrix with coefficient c and the 2×2 identity matrix by I, we must check that

$$B^2 = -I, C^2 = -I, K^2 = -I,$$
$$CK = B, KB = C, CB = -K, KC = -B, \text{ and } BK = -C.$$

c. We should check that ϕ is one to one.

SECTION 25

1. $a < x < x^2 < x^3 < \cdots < x^n \cdots$ for any $a \in R$.
3. $m + n\sqrt{2}$ is positive if $m > 0$ and $n < 0$, or if $m > 0$ and $m^2 > 2n^2$, or if $n < 0$ and $2n^2 > m^2$.
5. **i.** a c e d b **ii.** e c b a d
7. **i.** d a b c e **ii.** d c e a b
9. **i.** c a e d b **ii.** e c b a d
11. d b a e c **13.** d e b a c
15. **a.** T **c.** F **e.** T **g.** T **i.** F

SECTION 26

1. There are just nine possibilities:
$\phi(1, 0) = (1, 0)$ while $\phi(0, 1) = (0, 0)$ or $(0, 1)$,
$\phi(1, 0) = (0, 1)$ while $\phi(0, 1) = (0, 0)$ or $(1, 0)$,
$\phi(1, 0) = (1, 1)$ while $\phi(0, 1) = (0, 0)$, and
$\phi(1, 0) = (0, 0)$ while $\phi(0, 1) = (0, 0), (1, 0), (0, 1)$, or $(1, 1)$.
3. $\langle 0 \rangle = \{0\}, \mathbb{Z}_{12}/\{0\} \simeq \mathbb{Z}_{12}$
$\langle 1 \rangle = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11\}, \mathbb{Z}_{12}/\langle 1 \rangle \simeq \{0\}$
$\langle 2 \rangle = \{0, 2, 4, 6, 8, 10\}, \mathbb{Z}_{12}/\langle 2 \rangle \simeq \mathbb{Z}_2$
$\langle 3 \rangle = \{0, 3, 6, 9\}, \mathbb{Z}_{12}/\langle 3 \rangle \simeq \mathbb{Z}_3$
$\langle 4 \rangle = \{0, 4, 8\}, \mathbb{Z}_{12}/\langle 4 \rangle \simeq \mathbb{Z}_4$
$\langle 6 \rangle = \{0, 6\}, \mathbb{Z}_{12}/\langle 6 \rangle \simeq \mathbb{Z}_6$
9. Let $\phi : \mathbb{Z} \to \mathbb{Z} \times \mathbb{Z}$ be given by $\phi(n) = (n, 0)$ for $n \in \mathbb{Z}$.

11. R/R and $R/\{0\}$ are not of real interest because R/R is the ring containing only the zero element, and $R/\{0\}$ is isomorphic to R.
13. $\mathbb{Z}$ is an integral domain. $\mathbb{Z}/4\mathbb{Z}$ is isomorphic to $\mathbb{Z}_4$, which has a divisor 2 of 0.
15. $\{(n, n) \mid n \in \mathbb{Z}\}$. (Other answers are possible.)
31. The nilradical of $\mathbb{Z}_{12}$ is $\{0, 6\}$. The nilradical of $\mathbb{Z}$ is $\{0\}$ and the nilradical of $\mathbb{Z}_{32}$ is $\{0, 2, 4, 6, 8, \cdots, 30\}$.
35. **a.** Let $R = \mathbb{Z}$ and let $N = 4\mathbb{Z}$. Then $\sqrt{N} = 2\mathbb{Z} \neq 4\mathbb{Z}$
b. Let $R = \mathbb{Z}$ and let $N = 2\mathbb{Z}$. Then $\sqrt{N} = N$.

SECTION 27

1. $\{0, 2, 4\}$ and $\{0, 3\}$ are both prime and maximal.
3. $\{(0, 0), (1, 0)\}$ and $\{(0, 0), (0, 1)\}$ are both prime and maximal.
5. 1 **7.** 2 **9.** 1, 4 **15.** $2\mathbb{Z} \times \mathbb{Z}$ **17.** $4\mathbb{Z} \times \{0\}$
19. Yes. $x^2 - 6x + 6$ is irreducible over $\mathbb{Q}$ by Eisenstein with $p = 2$.
21. Yes. $\mathbb{Z}_2 \times \mathbb{Z}_3$
23. No. Enlarging the domain to a field of quotients, you would have to have a field containing two different prime fields $\mathbb{Z}_p$ and $\mathbb{Z}_q$, which is impossible.

SECTION 28

1. $-3x^3 + 7x^2y^2z - 5x^2yz^3 + 2xy^3z^5$
3. $2x^2yz^2 - 2xy^2z^2 - 7x + 3y + 10z^3$
5. $2z^5y^3x - 5z^3yx^2 + 7zy^2x^2 - 3x^3$
7. $10z^3 - 2z^2y^2x + 2z^2yx^2 + 3y - 7x$
9. $1 < z < y < x < z^2 < yz < y^2 < xz < xy < x^2 < z^3 < yz^2 < y^2z < y^3 < xz^2 < xyz < xy^2 < x^2z < x^2y < x^3 < \cdots$
11. $3y^2z^5 - 8z^7 + 5y^3z^3 - 4x$ **13.** $3yz^3 - 8xy - 4xz + 2yz + 38$
15. $\langle y^5 + y^3, y^3 + z, x - y^4 \rangle$ **17.** $\langle y^2z^3 + 3, -3y - 2z, y^2z^2 + 3 \rangle$
19. $\{1\}$ **21.** $\{x - 1\}$
23. $\{2x + y - 5, y^2 - 9y + 18\}$
The algebraic variety is $\{(1, 3), (-\frac{1}{2}, 6)\}$.
25. $\{x + y, y^3 - y + 1\}$
The algebraic variety consists of one point $(a, -a)$ where $a \approx 1.3247$.
27. **a.** T **c.** T **e.** T **g.** T **i.** F

SECTION 29

1. $x^2 - 2x - 1$ **3.** $x^2 - 2x + 2$
5. $x^{12} + 3x^8 - 4x^6 + 3x^4 + 12x^2 + 5$
7. $\text{Irr}(\alpha, \mathbb{Q}) = x^4 - \frac{2}{3}x^2 - \frac{62}{9}$; $\deg(\alpha, \mathbb{Q}) = 4$
9. Algebraic, $\deg(\alpha, F) = 2$
11. Transcendental
13. Algebraic, $\deg(\alpha, F) = 2$
15. Algebraic, $\deg(\alpha, F) = 1$
17. $x^2 + x + 1 = (x - \alpha)(x + 1 + \alpha)$
23. **a.** T **c.** T **e.** F **g.** F **i.** F
25. **b.** $x^3 + x^2 + 1 = (x - \alpha)(x - \alpha^2)[x - (1 + \alpha + \alpha^2)]$
27. It is the monic polynomial in $F[x]$ of *minimal* degree having α as a zero.

SECTION 30

1. $\{(0, 1), (1, 0)\}, \{(1, 1), (-1, 1)\}, \{(2, 1), (1, 2)\}$. (Other answers are possible.)
3. No. $2(-1, 1, 2) - 4(2, -3, 1) + (10, -14, 0) = (0, 0, 0)$
5. $\{1\}$ **7.** $\{1, i\}$ **9.** $\{1, \sqrt[4]{2}, \sqrt{2}, (\sqrt[4]{2})^3\}$
15. **a.** T **c.** T **e.** F **g.** F **i.** T
17. **a.** The **subspace of** V **generated by** S is the intersection of all subspaces of V containing S.
19. *Partial answer:* A basis for F^n is

$$\{(1, 0, \cdots, 0), (0, 1, \cdots, 0), \cdots, (0, 0, \cdots, 1)\}$$

where 1 is the multiplicative identity of F.
25. **a.** A homomorphism
b. *Partial answer:* The **kernel** (or **nullspace**) of ϕ is $\{\alpha \in V \mid \phi(\alpha) = 0\}$.
c. ϕ is an isomorphism of V with V' if $\text{Ker}(\phi) = \{0\}$ and ϕ maps V onto V'.

SECTION 31

1. $2, \{1, \sqrt{2}\}$ **3.** $4, \{1, \sqrt{3}, \sqrt{2}, \sqrt{6}\}$
5. $6, \{1, \sqrt{2}, \sqrt[3]{2}, \sqrt{2}(\sqrt[3]{2}), (\sqrt[3]{2})^2, \sqrt{2}(\sqrt[3]{2})^2\}$ **7.** $2, \{1, \sqrt{6}\}$
9. $9, \{1, \sqrt[3]{2}, \sqrt[3]{4}, \sqrt[3]{3}, \sqrt[3]{6}, \sqrt[3]{12}, \sqrt[3]{9}, \sqrt[3]{18}, \sqrt[3]{36}\}$
11. $2, \{1, \sqrt{2}\}$ **13.** $2, \{1, \sqrt{2}\}$
19. **a.** F **c.** F **e.** F **g.** F **i.** F
23. *Partial answer:* Extensions of degree 2^n for $n \in \mathbb{Z}^+$ are obtained.

SECTION 32

All odd-numbered answers require proofs and are not listed here.

SECTION 33

1. Yes **3.** Yes **5.** 6 **7.** 0

SECTION 34

1. **a.** $K = \{0, 3, 6, 9\}$.
b. $0 + K = \{0, 3, 6, 9\}, 1 + K = \{1, 4, 7, 10\}, 2 + K = \{2, 5, 8, 11\}$.
c. $\mu(0 + K) = 0, \mu(1 + K) = 2, \mu(2 + K) = 1$.
3. **a.** $HN = \{0, 2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22\}, H \cap N = \{0, 12\}$.
b. $0 + N = \{0, 6, 12, 18\}, 2 + N = \{2, 8, 14, 20\}, 4 + N = \{4, 10, 16, 22\}$.
c. $0 + (H \cap N) = \{0, 12\}, 4 + (H \cap N) = \{4, 16\}, 8 + (H \cap N) = \{8, 20\}$.
d. $\phi(0 + N) = 0 + (H \cap N), \phi(2 + N) = 8 + (H \cap N), \phi(4 + N) = 4 + (H \cap N)$.
5. **a.** $0 + H = \{0, 4, 8, 12, 16, 20\}, 1 + H = \{1, 5, 9, 13, 17, 21\}$,
$2 + H = \{2, 6, 10, 14, 18, 22\}, 3 + H = \{3, 7, 11, 15, 19, 23\}$.
b. $0 + K = \{0, 8, 16\}, 1 + K = \{1, 9, 17\}, 2 + K = \{2, 10, 18\}$,
$3 + K = \{3, 11, 19\}$,
$4 + K = \{4, 12, 20\}, 5 + K = \{5, 13, 21\}, 6 + K = \{6, 14, 22\}$,
$7 + K = \{7, 15, 23\}$.

c. $0 + K = \{0, 8, 16\}, 4 + K = \{4, 12, 20\}$.
d. $(0 + K) + (H/K) = H/K = \{0 + K, 4 + K\} = \{\{0, 8, 16\}, 4, 12, 20\}\}$
$(1 + K) + (H/K) = \{1 + K, 5 + K\} = \{\{1, 9, 17\}, \{5, 13, 21\}\}$
$(2 + K) + (H/K) = \{2 + K, 6 + K\} = \{\{2, 10, 18\}, \{6, 14, 22\}\}$
$(3 + K) + (H/K) = \{3 + K, 7 + K\} = \{\{3, 11, 19\}, \{7, 15, 23\}\}$.
e. $\phi(0 + H) = (0 + K) + (H/K), \phi(1 + H) = (1 + K) + (H/K)$,
$\phi(2 + H) = (2 + K) + (H/K), \phi(3 + H) = (3 + K) + (H/K)$.

SECTION 35

1. The refinements $\{0\} < 250\mathbb{Z} < 10\mathbb{Z} < \mathbb{Z}$ of $\{0\} < 10\mathbb{Z} < \mathbb{Z}$ and $\{0\} < 250\mathbb{Z} < 25\mathbb{Z} < \mathbb{Z}$ of $0 < 25\mathbb{Z} < \mathbb{Z}$ are isomorphic.
3. The given series are isomorphic.
5. The refinements
$\{(0, 0)\} < (4800\mathbb{Z}) \times \mathbb{Z} < (240\mathbb{Z}) \times \mathbb{Z} < (60\mathbb{Z}) \times \mathbb{Z} < (10\mathbb{Z}) \times \mathbb{Z} < \mathbb{Z} \times \mathbb{Z}$ of the first series and
$\{(0, 0)\} < \mathbb{Z} \times (4800\mathbb{Z}) < \mathbb{Z} \times (480\mathbb{Z}) < \mathbb{Z} \times (80\mathbb{Z}) < \mathbb{Z} \times (20\mathbb{Z}) < \mathbb{Z} \times \mathbb{Z}$ of the second series are isomorphic refinements.
7. $\{0\} < \langle 16\rangle < \langle 8\rangle < \langle 4\rangle < \langle 2\rangle < \mathbb{Z}_{48}$
$\{0\} < \langle 24\rangle < \langle 8\rangle < \langle 4\rangle < \langle 2\rangle < \mathbb{Z}_{48}$
$\{0\} < \langle 24\rangle < \langle 12\rangle < \langle 4\rangle < \langle 2\rangle < \mathbb{Z}_{48}$
$\{0\} < \langle 24\rangle < \langle 12\rangle < \langle 6\rangle < \langle 2\rangle < \mathbb{Z}_{48}$
$\{0\} < \langle 24\rangle < \langle 12\rangle < \langle 6\rangle < \langle 3\rangle < \mathbb{Z}_{48}$
9. $\{(\rho_0, 0)\} < A_3 \times \{0\} < S_3 \times (0) < S_3 \times \mathbb{Z}_2$
$\{(\rho_0, 0)\} < \{\rho_0\} \times \mathbb{Z}_2 < A_3 \times \mathbb{Z}_2 < S_3 \times \mathbb{Z}_2$
$\{(\rho_0, 0)\} < A_3 \times \{0\} < A_3 \times \mathbb{Z}_2 < S_3 \times \mathbb{Z}_2$
11. $\{\rho_0\} \times \mathbb{Z}_4$ **13.** $\{\rho_0\} \times \mathbb{Z}_4 \leq \{\rho_0\} \times \mathbb{Z}_4 \leq \{\rho_0\} \times \mathbb{Z}_4 \leq \cdots$.
17. **a.** *T* **c.** *T* **e.** *F* **g.** *F* **i.** *T*
i. The Jordan-Hölder theorem applied to the groups $\mathbb{Z}_n$ implies the Fundamental Theorem of Arithmetic.
19. Yes. $\{\rho_0\} < \{\rho_0, \rho_2\} < \{\rho_0, \rho_1, \rho_2, \rho_3\} < D_4$ is a composition (actually a principal) series and all factor groups are isomorphic to $\mathbb{Z}_2$ and are thus abelian.
21. *Chain (3)*

$$\begin{aligned}\{0\} &\leq \langle 12\rangle \leq \langle 12\rangle \leq \langle 12\rangle \\ &\leq \langle 12\rangle \leq \langle 12\rangle \leq \langle 4\rangle \\ &\leq \langle 2\rangle \leq \mathbb{Z}_{24} \leq \mathbb{Z}_{24}\end{aligned}$$

Chain (4)

$$\begin{aligned}\{0\} &\leq \langle 12\rangle < \langle 12\rangle \leq \langle 6\rangle \\ &\leq \langle 6\rangle \leq \langle 6\rangle \leq \langle 3\rangle \\ &\leq \langle 3\rangle \leq \mathbb{Z}_{24} \leq \mathbb{Z}_{24}\end{aligned}$$

Isomorphisms

$$\begin{aligned}&\langle 12\rangle/\{0\} \simeq \langle 12\rangle/\{0\} \simeq \mathbb{Z}_2, \quad \langle 12\rangle/\langle 12\rangle \simeq \langle 6\rangle/\langle 6\rangle \simeq \{0\}, \\ &\langle 12\rangle/\langle 12\rangle \simeq \langle 3\rangle/\langle 3\rangle \simeq \{0\}, \quad \langle 12\rangle/\langle 12\rangle \simeq \langle 12\rangle/\langle 12\rangle \simeq \{0\}, \\ &\langle 12\rangle/\langle 12\rangle \simeq \langle 6\rangle/\langle 6\rangle \simeq \{0\}, \quad \langle 4\rangle/\langle 12\rangle \simeq \mathbb{Z}_{24}/\langle 3\rangle \simeq \mathbb{Z}_3 \\ &\langle 2\rangle/\langle 4\rangle \simeq \langle 6\rangle/\langle 12\rangle \simeq \mathbb{Z}_2, \quad \mathbb{Z}_{24}/\langle 2\rangle \simeq \langle 3\rangle/\langle 6\rangle \simeq \mathbb{Z}_2 \\ &\mathbb{Z}_{24}/\mathbb{Z}_{24} \simeq \mathbb{Z}_{24}/\mathbb{Z}_{24} \simeq \{0\}\end{aligned}$$

SECTION 36

1. 3 **3.** 1, 3
5. The Sylow 3-subgroups are $\langle(1, 2, 3)\rangle$, $\langle(1, 2, 4)\rangle$, $\langle(1, 3, 4)\rangle$, and $\langle(2, 3, 4)\rangle$. Also $(3, 4)\langle(1, 2, 3)\rangle(3, 4) = \langle(1, 2, 4)\rangle$, etc.

SECTION 37

1. **a.** The conjugate classes are $\{\rho_0\}, \{\rho_2\}, \{\rho_1, \rho_3\}, \{\mu_1, \mu_2\}, \{\delta_1, \delta_2\}$.
b. $8 = 2 + 2 + 2 + 2$

3. **a.** T **c.** F **e.** T **g.** T **i.** F
e. This is somewhat a matter of opinion.

9. $24 = 1 + 6 + 3 + 8 + 6$

SECTION 38

1. $\{(1, 1, 1), (1, 2, 1), (1, 1, 2)\}$

3. No. $n(2, 1) + m(4, 1)$ can never yield an odd number for first coordinate.

7. $2\mathbb{Z} < \mathbb{Z}$, rank $r = 1$

SECTION 39

1. **a.** $a^2b^2a^3c^3b^{-2}, b^2c^{-3}a^{-3}b^{-2}a^{-2}$ **b.** $a^{-1}b^3a^4c^6a^{-1}, ac^{-6}a^{-4}b^{-3}a$

3. **a.** 16 **b.** 36 **c.** 36

5. **a.** 16 **b.** 36 **c.** 18

11. **a.** *Partial answer:* $\{1\}$ is a basis for $\mathbb{Z}_4$. **c.** Yes

13. **c.** A blop group on S is isomorphic to the *free group* $F[S]$ on S.

SECTION 40

1. $(a : a^4 = 1)$; $(a, b : a^4 = 1, b = a^2)$; $(a, b, c : a = 1, b^4 = 1, c = 1)$. (Other answers are possible.)

3. *Octic group:*

	1	a	a^2	a^3	b	ab	a^2b	a^3b
1	1	a	a^2	a^3	b	ab	a^2b	a^3b
a	a	a^2	a^3	1	ab	a^2b	a^3b	b
a^2	a^2	a^3	1	a	a^2b	a^3b	b	ab
a^3	a^3	1	a	a^2	a^3b	b	ab	a^2b
b	b	a^3b	a^2b	ab	1	a^3	a^2	a
ab	ab	b	a^3b	a^2b	a	1	a^3	a^2
a^2b	a^2b	ab	b	a^3b	a^2	a	1	a^3
a^3b	a^3b	a^2b	ab	b	a^3	a^2	a	1

Quaternion group: The same as the table for the octic group except that the 16 entries in the lower right corner are

a^2	a	1	a^3
a^3	a^2	a	1
1	a^3	a^2	a
a	1	a^3	a^2

5. $\mathbb{Z}_{21}$. $(a, b : a^7 = 1, b^3 = 1, ba = a^2b)$

SECTION 41

1. **a.** $2P_1P_3 - 3P_1P_4 + P_1P_6 - 3P_2P_3 + 3P_2P_4 - 5P_3P_4 + 4P_3P_6 - 5P_4P_6$
   b. No **c.** Yes
3. $C_i(P) = Z_i(P) = B_i(P) = H_i(P) = 0$ for $i > 0$. $B_0(P) = 0$. $Z_0(P) \simeq \mathbb{Z}$ and is generated by the 0-cycle P. $H_0(P) \simeq \mathbb{Z}$.
5. $C_i(X) = Z_i(X) = B_i(X) = H_i(X) = 0$ for $i > 0$. $B_0(X) \simeq \mathbb{Z}$ and is generated by the 0-chain $P_2 - P_1$. $Z_0(X) \simeq \mathbb{Z} \times \mathbb{Z}$ and is generated by the two 0-cycles P_1 and P_2. Since $Z_0(X)/B_0(X)$ "indentifies P_1 with P_2," $H_0(X) \simeq \mathbb{Z}$ and is generated by the coset $P_1 + B_0(X)$.
7. **a.** An **oriented n-simplex** is an ordered sequance $P_1P_2 \cdots P_{n+1}$.
   b. The **boundary of** $P_1P_2 \cdots P_{n+1}$ is given by

$$\partial_n(P_1P_2 \cdots P_{n+1}) \sum_{i=1}^{n+1} (-1)^{i+1} P_1P_2 \cdots P_{i-1}P_{i+1} \cdots P_{n+1}.$$

   c. Each individual *summand* of the boundary of an oriented n-simplex is a **face of the simplex.**
11. **a.** $\delta^{(n)}\left(\sum_i m_i\sigma_i\right) = \sum_i m_i\delta^{(n)}(\sigma_i)$
13. $H^{(n)}(X) = Z^{(n)}(X)/B^{(n)}(X)$
    $H^{(0)}(S) \simeq \mathbb{Z}$ and is generated by $(P_1 + P_2 + P_3 + P_4) + \{0\}$
    $H^{(1)}(S) = 0$
    $H^{(2)}(S) \simeq \mathbb{Z}$ and is generated by $P_1P_2P_3 + B^{(2)}(S)$

SECTION 42

1. $H_0(X) \simeq \mathbb{Z}$. $H_1(X) \simeq \mathbb{Z} \times \mathbb{Z}$. $H_n(X) = 0$ for $n > 1$.
3. $H_0(X) \simeq \mathbb{Z} \times \mathbb{Z}$. $H_1(X) \simeq \mathbb{Z}$. $H_2(X) \simeq \mathbb{Z}$. $H_n(X) = 0$ for $n > 2$.
5. $H_0(X) \simeq \mathbb{Z}$. $H_1(X) \simeq \mathbb{Z}$. $H_2(X) \simeq \mathbb{Z}$. $H_n(X) = 0$ for $n > 2$.
7. **a.** T **c.** F **e.** T **g.** T **i.** F
9. $H_0(X) \simeq \mathbb{Z}$. $H_1(X) \simeq \mathbb{Z} \times \mathbb{Z} \times \mathbb{Z}$. $H_2(X) \simeq \mathbb{Z} \times \mathbb{Z}$. $H_n(X) = 0$ for $n > 2$.
11. $H_0(X) \simeq \mathbb{Z}$. $H_1(X) \simeq \mathbb{Z} \times \mathbb{Z} \times \mathbb{Z} \times \mathbb{Z}$. $H_2(X) \simeq \mathbb{Z}$. $H_n(X) = 0$ for $n > 2$.

SECTION 43

1. Both counts show that $\chi(X) = 1$.
3. It will hold for a square region, for such a region is homeomorphic to E^2. It obviously does not hold for two disjoint 2-cells, for each can be mapped continuously onto the other, and such a map has no fixed points.
5. $H_0(X) \simeq \mathbb{Z} \times \mathbb{Z}$. $H_1(X) \simeq \mathbb{Z} \times \mathbb{Z} \times \mathbb{Z}_2 \times \mathbb{Z}_2$. $H_n(X) = 0$ for $n > 1$.
7. $2 - 2n$

9. $H_0(X) \simeq \mathbb{Z}$. $H_1(X) \simeq \underbrace{\mathbb{Z} \times \mathbb{Z} \times \cdots \times \mathbb{Z}}_{(q-1)\text{ factors}} \times \mathbb{Z}_2$. $H_n(X) = 0$ for $n > 1$.

11. Let Q be a vertex of b, and let c be the 2-chain consisting of all 2-simplexes of X, all oriented the same way, so that $c \in Z_2(X)$.

a. f_{*0} is given by $f_{*0}(Q + B_0(X)) = Q + B_0(b)$.
f_{*1} is given by $f_{*1}((ma + nb) + B_1(X)) = nb + B_1(b)$.
f_{*2} is given by $f_{*2}(c + B_1(X)) = 0$.

b. f_{*0} is as in (a).
f_{*1} is given by $f_{*1}((ma + nb) + B_1(X)) = 2nb + B_1(b)$.
f_{*2} is as in (a).

13. Let Q be a vertex on b.
f_{*0} is given by $f_{*0}(Q + B_0(X)) = Q + B_0(b)$.
f_{*1} is given by $f_{*1}((ma + nb) + B_1(X)) = nb + B_1(b)$, where $m = 0, 1$.
f_{*2} is trivial, since both $H_2(X)$ and $H_2(b)$ are 0.

SECTION 44

5. For Theorem 44.4, the condition $f_{k-1}\partial_k = \partial'_k f_k$ implies that

$$f_{k-1}(B_{k-1}(A)) \subseteq B_{k-1}(A').$$

Then Exercise 14.39 shows that f_{k-1} induces a natural homomorphism of $Z_{k-1}(A)/B_{k-1}(A)$ into $Z_{k-1}(A')/B_{k-1}(A')$. This is the correct way to view Theorem 44.4.
For Theorem 44.7, if we use Exercise 14.39, the fact that $\partial_k(A'_k) \subseteq A'_{k-1}$ shows that ∂_k induces a natural homomorphism $\bar{\partial}_k : (A_k/A'_k) \to (A_{k-1}/A'_{k-1})$.

7. The exact homology sequence is

$$\begin{aligned}[H_2(a) = 0] &\xrightarrow{i_{*2}} [H_2(X) \simeq \mathbb{Z}] \xrightarrow{j_{*2}} [H_2(X, a) \simeq \mathbb{Z}] \xrightarrow{\partial_{*2}} [H_1(a) \simeq \mathbb{Z}] \\ &\xrightarrow{i_{*1}} [H_1(X) \simeq \mathbb{Z} \times \mathbb{Z}] \xrightarrow{j_{*1}} [H_1(X, a) \simeq \mathbb{Z}] \xrightarrow{\partial_{*1}} [H_0(a) \simeq \mathbb{Z}] \\ &\xrightarrow{i_{*0}} [H_0(X) \simeq \mathbb{Z}] \xrightarrow{j_{*0}} [H_0(X, a) = 0].\end{aligned}$$

j_{*2} maps a generator $c + B_2(X)$ of $H_2(X)$ onto the generator

$$(c + C_2(a)) + B_2(X, a)$$

of $H_2(X, a)$ and is an isomorphism. Thus (kernel j_{*2}) = (image i_{*2}) = 0.
∂_{*2} maps everything onto 0, so (kernal ∂_{*2}) = (image j_{*2}) $\simeq \mathbb{Z}$.
i_{*1} maps the generator $a + B_1(a)$ onto $(a + 0b) + B_1(X)$, so i_{*1} is an isomorphism *into*, and (kernal i_{*1}) = (image ∂_{*2}) = 0.
j_{*1} maps $(ma + nb) + B_1(X)$ onto $(nb + C_1(a)) + B_1(X, a)$, so (kernal j_{*1}) = (image i_{*1}) $\simeq \mathbb{Z}$.
∂_{*1} maps $(nb + C_1(a)) + B_1(X, a)$ onto 0, so (kernal ∂_{*1}) = (image j_{*1}) $\simeq \mathbb{Z}$.
For a vertex Q of a, i_{*0} maps $Q + B_0(a)$ onto $Q + B_0(X)$, so i_{*0} is an isomorphism, and (kernal i_{*0}) = (image ∂_{*1}) = 0.
j_{*0} maps $Q + B_0(X)$ onto $B_0(X, a)$ in $H_0(X, a)$, so (kernal j_{*0}) = (image i_{*0}) $\simeq \mathbb{Z}$.

9. The answer is formally identical with that in Exercise 44.7.

11. *Partial answer:* The exact homology sequence is

$$\begin{aligned}[H_2(Y) = 0] &\xrightarrow{i_{*2}} [H_2(X) = 0] \xrightarrow{j_{*2}} [H_2(X, Y) \simeq \mathbb{Z}] \xrightarrow{\partial_{*2}} [H_1(Y) \simeq \mathbb{Z} \times \mathbb{Z}] \\ &\xrightarrow{i_{*1}} [H_1(X) \simeq \mathbb{Z}] \xrightarrow{j_{*1}} [H_1(X, Y) \simeq \mathbb{Z}] \xrightarrow{\partial_{*1}} [H_0(Y) \simeq \mathbb{Z} \times \mathbb{Z}] \\ &\xrightarrow{i_{*0}} [H_0(X) \simeq \mathbb{Z}] \xrightarrow{j_{*0}} [H_0(X, Y) = 0].\end{aligned}$$

The verification of exactness is left to you. Note that the edge P_1Q_1 of Fig. 42.11 gives rise to a generator of $H_1(X, Y)$. Starting with ∂_{*2}, these maps are very interesting.

SECTION 45

1. Yes **3.** No **5.** No. **7.** Yes

9. In $\mathbb{Z}[x]$: only $2x - 7, -2x + 7$

In $\mathbb{Q}[x]$: $4x - 14, x - \frac{7}{2}, 6x - 21, -8x + 28$

In $\mathbb{Z}_{11}[x]$: $2x - 7, 10x - 2, 6x + 1, 3x - 5, 5x - 1$

11. $26, -26$ **13.** $198, -198$

15. It is already "primitive" because every nonzero element of $\mathbb{Q}$ is a unit. Indeed $18ax^2 - 12ax + 48a$ is primitive for all $a \in \mathbb{Q}, a \neq 0$.

17. $2ax^2 - 3ax + 6a$ is primitive for all $a \neq 0$ in $\mathbb{Z}_7$ because every such element a is a unit in $\mathbb{Z}_7$.

21. **a.** T **c.** T **e.** T **g.** F **i.** F

i. Either p or one of its associates must appear in every factorization *into irreducibles.*

23. $2x + 4$ is irreducible in $\mathbb{Q}[x]$ but not in $\mathbb{Z}[x]$.

31. *Partial answer:* $x^3 - y^3 = (x - y)(x^2 + xy + y^2)$

SECTION 46

1. Yes **3.** No. (1) is violated. **5.** Yes

7. 61 **9.** $x^3 + 2x - 1$ **11.** 66

13. **a.** T **c.** T **e.** T **g.** T **i.** T

23. *Partial answer:* The equation $ax = b$ has a solution in $\mathbb{Z}_n$ for nonzero $a, b \in \mathbb{Z}_n$ if and only if the positive gcd of a and n in $\mathbb{Z}$ divides b.

SECTION 47

1. $5 = (1 + 2i)(1 - 2i)$ **3.** $4 + 3i = (1 + 2i)(2 - i)$

5. $6 = (2)(3) = (-1 + \sqrt{-5})(-1 - \sqrt{-5})$ **7.** $7 - i$

15. **c.** **i)** order 9, characteristic 3 **ii)** order 2, characteristic 2

iii) order 5, characteristic 5

SECTION 48

1. $\sqrt{2}, -\sqrt{2}$ **3.** $3 + \sqrt{2}, 3 - \sqrt{2}$ **5.** $\sqrt{2} + i, \sqrt{2} - i, -\sqrt{2} + i, -\sqrt{2} - i$

7. $\sqrt{1 + \sqrt{2}}, -\sqrt{1 + \sqrt{2}}, \sqrt{1 - \sqrt{2}}, -\sqrt{1 - \sqrt{2}}$ **9.** $\sqrt{3}$

11. $-\sqrt{2} + 3\sqrt{5}$ **13.** $-\sqrt{2} + \sqrt{45}$

15. **a.** $\mathbb{Q}$ **b.** $\mathbb{Q}(\sqrt{6})$ **c.** $\mathbb{Q}$

17. $\mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{5})$ **19.** $\mathbb{Q}(\sqrt{3}, \sqrt{10})$ **21.** $\mathbb{Q}$

25. **a.** $3 - \sqrt{2}$ **b.** They are the same maps.

27. $\sigma_3(0) = 0, \sigma_3(1) = 1, \sigma_3(2) = 2, \sigma_3(\alpha) = -\alpha, \sigma_3(2\alpha) = -2\alpha,$
$\sigma_3(1 + \alpha) = 1 - \alpha, \sigma_3(1 + 2\alpha) = 1 - 2\alpha, \sigma_3(2 + \alpha) = 2 - \alpha,$
$\sigma_3(2 + 2\alpha) = 2 - 2\alpha; \mathbb{Z}_3(\alpha)_{\{\sigma 3\}} = \mathbb{Z}_3$

29. **a.** F **c.** T **e.** F **g.** T **i.** T

37. Yes

SECTION 49

1. The identity map of E onto E;
τ given by $\tau(\sqrt{2}) = \sqrt{2}, \tau(\sqrt{3}) = -\sqrt{3}, \tau(\sqrt{5}) = -\sqrt{5}$

3. τ_1 given by $\tau_1(\sqrt{2}) = \sqrt{2}, \tau_1(\sqrt{3}) = \sqrt{3}, \tau_1(\sqrt{5}) = -\sqrt{5}$;
τ_2 given by $\tau_2(\sqrt{2}) = \sqrt{2}, \tau_2(\sqrt{3}) = -\sqrt{3}, \tau_2(\sqrt{5}) = \sqrt{5}$;
τ_3 given by $\tau_3(\sqrt{2}) = -\sqrt{2}, \tau_3(\sqrt{3}) = \sqrt{3}, \tau_3(\sqrt{5}) = \sqrt{5}$;
τ_4 given by $\tau_4(\sqrt{2}) = -\sqrt{2}, \tau_4(\sqrt{3}) = -\sqrt{3}, \tau_4(\sqrt{5}) = -\sqrt{5}$

5. The identity map of $\mathbb{Q}(\sqrt[3]{2}, \sqrt{3})$ into itself;
τ_1 given by $\tau_1(\alpha_1) = \alpha_1, \tau_1(\sqrt{3}) = -\sqrt{3}$ where $\alpha_1 = \sqrt[3]{2}$;
τ_2 given by $\tau_2(\alpha_1) = \alpha_2, \tau_2(\sqrt{3}) = \sqrt{3}$ where $\alpha_2 = \sqrt[3]{2}(-1 + i\sqrt{3})/2$;
τ_3 given by $\tau_3(\alpha_1) = \alpha_2, \tau_3(\sqrt{3}) = -\sqrt{3}$;
τ_4 given by $\tau_4(\alpha_1) = \alpha_3, \tau_4(\sqrt{3}) = \sqrt{3}$ where $\alpha_3 = \sqrt[3]{2}(-1 - i\sqrt{3})/2$;
τ_5 given by $\tau_5(\alpha_1) = \alpha_3, \tau_5(\sqrt{3}) = -\sqrt{3}$;

7. **a.** $\mathbb{Q}(\pi^2)$ **b.** τ_1 given by $\tau_1(\sqrt{\pi}) = i\sqrt{\pi}$, τ_2 given by $\tau_2(\sqrt{\pi}) = -i\sqrt{\pi}$

SECTION 50

1. 2 **3.** 4 **5.** 2 **7.** 1 **9.** 2 **13.** $1 \leq [E : F] \leq n!$

15. Let $F = \mathbb{Q}$ and $E = \mathbb{Q}(\sqrt{2})$. Then

$$f(x) = x^4 - 5x^2 + 6 = (x^2 - 2)(x^2 - 3)$$

has a zero in E, but does not split in E.

23. **a.** 6

SECTION 51

1. $\alpha = \sqrt[6]{2} = 2/(\sqrt[3]{2}\sqrt{2}).\sqrt{2} = (\sqrt[6]{2})^3, \sqrt[3]{2} = (\sqrt[6]{2})^2$. (Other answers are possible.)

3. $\alpha = \sqrt{2} + \sqrt{3}$. $\sqrt{2} = (\frac{1}{2})\alpha^3 - (\frac{9}{2})\alpha, \sqrt{3} = (\frac{11}{2})\alpha - (\frac{1}{2})\alpha^3$. (Other answers are possible.)

7. $f(x) = x^4 - 4x^2 + 4 = (x^2 - 2)^2$. Here $f(x)$ is not an irreducible polynomial. Every irreducible factor of $f(x)$ has zeros of multiplicty 1 only.

15. **b.** The field F **c.** $F[x^P]$

SECTION 52

1. $\mathbb{Z}_3(y^3, z^9)$ **3.** $\mathbb{Z}_3(y^4, z^2)$

5. **a.** F **c.** F **e.** F **g.** T **i.** T

SECTION 53

1. 8 **3.** 8 **5.** 4 **7.** 2

9. The group has two elements, the identity automorphism ι of $\mathbb{Q}(i)$ and σ such that $\sigma(i) = -i$.

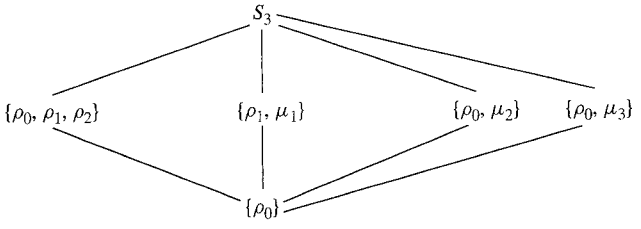
11. **a.** Let $\alpha_1 = \sqrt[3]{2}$, $\alpha_2 = \sqrt[3]{2}\dfrac{-1 + i\sqrt{3}}{2}$, and $\alpha_3 = \sqrt[3]{2}\dfrac{-1 - i\sqrt{3}}{2}$.

The maps are
ρ_0, where ρ_0 is the identity map;
ρ_1, where $\rho_1(\alpha_1) = \alpha_2$ and $\rho_1(i\sqrt{3}) = i\sqrt{3}$;

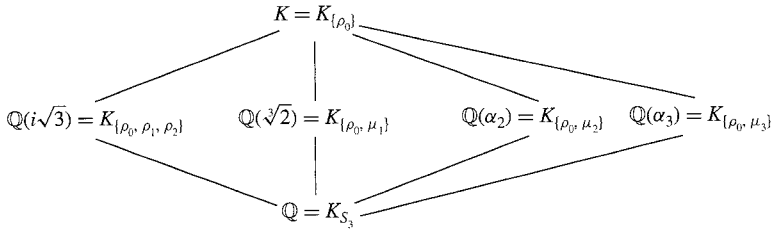
ρ_2, where $\rho_2(\alpha_1) = \alpha_3$ and $\rho_2(i\sqrt{3}) = i\sqrt{3}$;
μ_1, where $\mu_1(\alpha_1) = \alpha_1$ and $\mu_1(i\sqrt{3}) = -i\sqrt{3}$;
μ_2, where $\mu_2(\alpha_1) = \alpha_3$ and $\mu_2(i\sqrt{3}) = -i\sqrt{3}$;
μ_3, where $\mu_3(\alpha_1) = \alpha_2$ and $\mu_3(i\sqrt{3}) = -i\sqrt{3}$.

b. S_3. The notation in (a) was chosen to coincide with the notation for S_3 in Example 8.7.

c.



Group diagram



Field diagram

13. The splitting field of $(x^3 - 1) \in \mathbb{Q}[x]$ is $\mathbb{Q}(i\sqrt{3})$, and the group is cyclic of order 2 with elements: ι, where ι is the identity map of $\mathbb{Q}(i\sqrt{3})$, and σ, where $\sigma(i\sqrt{3}) = -i\sqrt{3}$.

15. **a.** *F* **c.** *T* **e.** *T* **g.** *F* **i.** *F*

25. *Partial answer:* $G(K/(E \vee L)) = G(K/E) \cap G(K/L)$

SECTION 54

3. $\mathbb{Q}(\sqrt[4]{2}, i)$: $\sqrt[4]{2} + i$, $x^8 + 4x^6 + 2x^4 + 28x^2 + 1$;
$\mathbb{Q}(\sqrt[4]{2})$: $\sqrt[4]{2}$, $x^4 - 2$;
$\mathbb{Q}(i\sqrt[4]{2}))$: $i(\sqrt[4]{2})$, $x^4 - 2$;
$\mathbb{Q}(\sqrt{2}, i)$: $\sqrt{2} + i$, $x^4 - 2x^2 + 9$;
$\mathbb{Q}(\sqrt[4]{2} + i(\sqrt[4]{2}))$: $\sqrt[4]{2} + i(\sqrt[4]{2})$, $x^4 + 8$;
$\mathbb{Q}(\sqrt[4]{2} - i(\sqrt[4]{2}))$: $\sqrt[4]{2} - i(\sqrt[4]{2})$, $x^4 + 8$;
$\mathbb{Q}(\sqrt{2})$: $\sqrt{2}$, $x^2 - 2$;
$\mathbb{Q}(i)$: i, $x^2 + 1$;
$\mathbb{Q}(i\sqrt{2})$: $i\sqrt{2}$, $x^2 + 2$;
$\mathbb{Q}$: 1, $x - 1$

5. The group is cyclic of order 5, and its elements are

	ι	σ_1	σ_2	σ_3	σ_4
$\sqrt[5]{2} \to$	$\sqrt[5]{2}$	$\zeta(\sqrt[5]{2})$	$\zeta^2(\sqrt[5]{2})$	$\zeta^3(\sqrt[5]{2})$	$\zeta^4(\sqrt[5]{2})$

where $\sqrt[5]{2}$ is the real 5th root of 2.

7. The splitting field of $x^8 - 1$ over $\mathbb{Q}$ is the same as the splitting field of $x^4 + 1$ over $\mathbb{Q}$, so a complete description is contained in Example 54.7. (This is the easiest way to answer the problem.)

9. **a.** $s_1{}^2 - 2s_2$ **b.** $\dfrac{s_1 s_2 - 3s_3}{s_3}$

SECTION 55

3. **a.** 16 **b.** 400 **c.** 2160

5. 3^0

7. $\Phi_3(x)$ over $\mathbb{Z}_2$ is $x^2 + x + 1$.
$\Phi_8(x)$ over $\mathbb{Z}_3$ is $x^4 + 1 = (x^2 + x + 2)(x^2 + 2x + 2)$.

9. **a.** T **c.** F **e.** T **g.** T **i.** F

11. $\Phi_1(x) = x - 1$
$\Phi_2(x) = x + 1$
$\Phi_3(x) = x^2 + x + 1$
$\Phi_4(x) = x^2 + 1$
$\Phi_5(x) = x^4 + x^3 + x^2 + x + 1$
$\Phi_6(x) = x^2 - x + 1$

SECTION 56

1. No. Yes, K is an extension of $\mathbb{Z}_2$ by radicals.

3. **a.** T **c.** T **e.** T **g.** T **i.** F ($x^3 - 2x$ over $\mathbb{Q}$ gives a counterexample.)

APPENDIX

1. $\begin{bmatrix} 2 & 1 \\ 2 & 7 \end{bmatrix}$ **3.** $\begin{bmatrix} -3 + 2i & -1 - 4i \\ 2 & -i \\ 0 & -i \end{bmatrix}$

5. $\begin{bmatrix} 5 & 16 & -3 \\ 0 & -18 & 24 \end{bmatrix}$ **7.** $\begin{bmatrix} 1 & -i \\ 4 - 6i & -2 - 2i \end{bmatrix}$

9. $\begin{bmatrix} 8 & -8i \\ 8i & 8 \end{bmatrix}$ **11.** $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ **13.** -48

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